United States Patent Application Publication	20250286304
Kind Code	A1
Publication Date	September 11, 2025
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Front-Facing Locking Feature for Quick Connect Terminal

Abstract

An electrical terminal with a contact portion having a bottom wall with resilient contact arms extending therefrom. The bottom wall has a base portion proximate a back end of the bottom wall. The back end is removed from a mating interface of the bottom wall. A locking latch extends from and is integrally attached to the base portion. The locking latch has a free end proximate the mating interface. The free end of the locking latch is moved in a direction perpendicular to an axis of insertion as a mating terminal is inserted into the contact portion of the terminal.

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Appl. No.: **18/599364**

Filed: **March 08, 2024**

Publication Classification

Int. Cl.: **H01R13/432** (20060101); **H01R13/115** (20060101); **H01R13/187** (20060101); **H01R101/00** (20060101)

U.S. Cl.:

CPC **H01R13/432** (20130101); **H01R13/115** (20130101); **H01R13/187** (20130101); H01R2101/00 (20130101)

Background/Summary

FIELD OF THE INVENTION

[0001] The present invention is directed to a quick connect electrical terminal. In particular, the invention is directed to a front facing locking arm which provides sufficient mechanical and electrical performance while decreasing the size of the terminal.

BACKGROUND OF THE INVENTION

[0002] Quick connect electrical terminals, such as tab receptacle terminals, which are adapted for quick make and break connections with a mating terminal or mating tab, are known. Terminals of this kind are used to make an electrical connection to a male or tab terminal which is inserted and held in the socket terminal.

[0003] Known terminals incorporate rear facing locking latches the deflect during insertion. The rear facing locking latches provide high retention forces to prevent the unwanted disengagement of mating connectors tabs from the terminals. However, various problems are encountered when using such a rear facing locking latches. Such rear facing locking latches generally have short beam lengths which can cause high stresses and plastic deformations in the latches when used. The rear facing locking latches also have a tendency to fail due to secondary bending of the latches along the locking latch length, resulting in the unwanted disengagement of the tabs.

[0004] The rear facing locking latches often have upward facing latch tips which allow a user to depress the rear facing locking latches to remove the mating tabs from the terminals. The upward facing latch tips of the rear facing locking latches are prone to excessive deflection during reeling, plating, shipping and crimping resulting in inconsistent retention forces in the rear facing locking latches. In addition, as the upward facing latch tips are stamped and formed form the terminals, large cutouts are provided in the transition areas of the terminals. This can result in the terminals having limited current carrying capacity and an increased overall length.

[0005] It would, therefore, be beneficial to provide a quick connect receptacle terminal which has a locking latch which does not have the disadvantages recited above. In particular, it would be beneficial to provide a front facing locking latch which has a longer effective beam length and which eliminates the need for an upwardly facing latch tip.

SUMMARY OF THE INVENTION

[0006] An embodiment is directed to an electrical terminal with a contact portion having a bottom wall with resilient contact arms extending therefrom. The bottom wall has a base portion proximate a back end of the bottom wall. The back end is removed from a mating interface of the bottom wall. A locking latch extends from and is integrally attached to the base portion. The locking latch has a free end proximate the mating interface. The free end of the locking latch is moved in a direction perpendicular to an axis of insertion as a mating terminal is inserted into the contact portion of the terminal.

[0007] An embodiment is directed to a quick disconnect electrical terminal. The quick disconnect electrical terminal includes a contact portion having a bottom wall with resilient contact arms extending therefrom. The bottom wall has a base portion proximate a back end of the bottom wall. The back end is removed from a mating interface of the bottom wall. The bottom wall has a first section, a locking latch and a third section. The locking latch extends from and is integrally attached to the base portion. The locking latch has a free end proximate the mating interface. The locking latch is separated from the first section by a first slot and separated from the third section by a second slot. The free end of the locking latch is moved independently of the first section and the third section in a direction perpendicular to an axis of insertion of a mating terminal as the mating terminal is inserted into the contact portion of the terminal.

[0008] Other features and advantages of the present invention will be apparent from the following more detailed description of the preferred embodiment, taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is perspective view of an illustrative embodiment of a receptacle terminal according to the present invention, a mating tab terminal is shown prior to being inserted into the receptacle terminal.

[0010] FIG. **2** is a top view of the receptacle terminal of FIG. **1**.

[0011] FIG. **3** is a bottom view of the receptacle terminal of FIG. **1**.

[0012] FIG. **4** is a front view of the receptacle terminal of FIG. **1** prior to the mating tab terminal being inserted into a terminal receiving area of the receptacle terminal.

[0013] FIG. **5** is a cross-sectional view of the receptacle terminal of FIG. **1** positioned in a connector housing, a mating tab is shown partially inserted into the housing and the receptacle terminal.

[0014] FIG. **6** is a cross-sectional view similar to that of FIG. **5**, with the mating tab fully inserted into the housing and the receptacle terminal.

[0015] FIG. **7** is a cross-sectional view similar to that of FIG. **5**, with a portion of the housing engaging a locking latch of the receptacle terminal to allow the mating tab to be released from the receptacle terminal and the housing.

DETAILED DESCRIPTION OF THE INVENTION

[0016] The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivative thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise.

[0017] Moreover, the features and benefits of the invention are illustrated by reference to the preferred embodiments. Accordingly, the invention expressly should not be limited to such embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features, the scope of the invention being defined by the claims appended hereto.

[0018] As best shown in FIGS. **1** through **4**, a quick disconnect electrical receptacle, socket or female terminal **10** includes a contact portion **12**, a wire barrel portion **14** behind the contact portion **12** and an insulation barrel portion **16** behind the wire barrel portion **14**. The wire barrel portion **14** is configured for a crimped connection with an end of a conductive core of an insulated wire. The insulation barrel portion **16** is configured for a crimped connection with an end of the insulation coating or jacket of the wire. Although a wire barrel portion **14** and an insulation barrel portion **16** are shown, the contact portion **12** can be used with other types of termination members without departing from the scope of the invention. In the illustrative embodiment shown, the terminal **10** is stamped and formed from a metal plate having good electrical conductivity.

[0019] The contact portion **12** includes a bottom wall **20** and resilient contact arms **22***a*, **22***b* which extend from either side of the bottom wall **20**. As shown in FIG. **3**, the bottom wall **20** has a first section **24**, a second section or locking latch **26** and a third section **28**. The first section **24**, the locking latch **26** and the third section **28** extend from, and are integrally or fixedly attached to, a base portion **30** of the bottom wall **20**. The base portion **30** is spaced from a mating interface **31** of the bottom wall **20** and is positioned proximate the back end **33** of the bottom wall **20**.

[0020] The first section **24** extends from base portion **30** to a free end **32**. The resilient contact arm **22***a* extends from the side of the first section **24**.

[0021] The second section or locking latch **26** extends from base portion **30** to a free end **34**. The locking latch **26** is separated from the first section **24** by a slot or gap **36**. The slot or gap **36** allows the free end **34** of the locking latch **26** to move independently relative to the free end **32** of the first section **24** in a direction perpendicular to the axis of insertion of a mating terminal **38** in the contact portion **12** of the terminal **10**. As shown in FIGS. **5** through **7**, the locking latch **26** has sloping mating section **37** provide proximate the free end **34**, a bent mounting section **39** extending from the base portion **30**, and a securing section **41** which extends between the mating section **37** and the mounting section **39**. The bent mounting section **41** positions the securing section **41** is a plane which is above and generally parallel to a plane of the base portion **30**.

[0022] The securing section **41** of the locking latch **26** has a projection or embossment **40**, such as, but not limited to, a projection, detent, dimple or lance which is formed from the locking latch **26** to create a raised area on an inner surface of the locking latch **26**. The projection or embossment **40** engages the mating terminal **38** as the mating terminal **38** is inserted into the terminal **10**.

[0023] The projection or embossment **40** is provided is spaced from where the second section or locking latch **26** extends from and is attached to the base portion **30** by a distance D**1**. The length D**1** is configured to provide a long effective beam length from the base **30** to the projection or embossment **40**, thereby reducing stress and plastic deformation of the second section or locking latch **26** as the mating terminal **38** is inserted into the terminal **10**. In the illustrative embodiment, the distance D**1** is configured to provide an effective beam length which prevents the locking latch **26** from taking a permanent set as the mating terminal **38** is inserted into the terminal **10**.

[0024] The third section **28** extends from base portion **30** to a free end **42**. The resilient contact arm **22***b* extends from the side of the third section **28**. The third section **28** is separated from the locking latch **26** by a slot or gap **44**. The slot or gap **44** allows the free end **42** of the third section **28** to move independently relative to the free end **34** of the locking latch **26** and relative to the free end **32** of the first section **24** in a direction perpendicular to the axis of insertion of the mating terminal **38** (FIGS. **1** and **5**) in the contact portion **12** of the terminal **10**.

[0025] The free end **32** of the first section **24**, the free end **34** of the locking latch **26** and the free end **42** of the third section **28** form the split beam mating interface **31**.

[0026] In the illustrative embodiment shown in FIG. **1**, each resilient arm **22***a*, **22***b* has a resilient contact section **46** with an arcuate or curled portion which extends from the bottom wall **20** to a mating terminal engaging member **48**. The configuration of the resilient contact arms **22***a*, **22***b* allows the stiffness and spring rate of the resilient contact arms **22***a*, **22***b* to be controlled.

[0027] The mating terminal engagement members **48** extend from the resilient contact section **46**. A mating terminal engagement surface **50** is provided on each mating terminal engaging member **48**. In the embodiment shown, the mating terminal engaging member **48** extends from the resilient contact arms **22***a*, **22***b*, positioning the mating terminal engagement surface **50** at the top of a terminal mating slot **52**. The configuration of the resilient contact arms **22***a*, **22***b* provides resiliency to allow the mating terminal engaging member **48** to move relative to the bottom wall **20** as the mating terminal is inserted into the mating slot **52**. The mating terminal engagement surfaces **50** have an arcuate or rounded configuration. However, other configurations of the engagement surfaces **50** may be used.

[0028] The mating terminal engagement members **48** have end surfaces **54** which are positioned proximate to each other, but are spaced apart to form a slot **56**. The width of the slot **56** is large enough to allow the end surfaces **54** and their respective contact arms **22***a*, **22***b* to move independently of each other. However, the width of the slot **56** is minimized to increase the beam length of the contact arms **22***a*, **22***b*. The beam length allows the contact arms **22***a*, **22***b* to have enhanced resilient or spring characteristics, allowing the contact arms **22***a*, **22***b* to move without taking a permanent set, failing or yielding. The slot **56** is positioned in line with the projection or embossment **40** of the second portion **26** of the bottom wall **20**.
[0029] As shown in FIG. **4**, in the initial, unstressed position, prior to the insertion of the mating terminal **38** into the mating slot **52**, the embossment **40** of the second section or locking latch **26** is positioned proximate to, but not in engagement with, the mating terminal engagement surfaces **50** of the mating terminal engagement members **48**. In this unstressed position, the resilient arm **22***a* and the resilient arm **22***b* are essentially mirror images of each other. The mating terminal engagement surfaces **50** of the mating terminal engagement members **48** are positioned in essentially the same plane which is transverse to the longitudinal axis of the terminal **10**.
[0030] In the unstressed position, the mating terminal engagement surfaces **50** of the mating terminal engagement members **48** and the second section or locking latch **26** define the mating slot **52**. In this position, the resilient arms **22***a*, **22***b* are in an unstressed position. The second section or locking latch **26** is also in an unstressed position. As previously described, the slot **36** and the slot **44** provided in the bottom wall **20** allow the second section or locking latch **26** to move independently of the first section **24** and the third section **28**.
[0031] As shown in FIGS. **5** through **7**, the terminals **10** are positioned in terminal receiving cavities **80** of housing **82**. The terminal receiving cavities **80** have openings **84** which extend through a front surface **86** of the housing **82**. Locking latch engaging surfaces **88** are provided in the terminal receiving cavities **80** proximate the openings **84**. In the embodiment shown, the locking latch engaging surfaces **88** have an arcuate configuration and extend in a direction which is essentially perpendicular to the longitudinal axes of the terminal receiving cavities **80**. However, other configurations of the housing **80** and the locking latch engaging surfaces **88** may be used.
[0032] Referring to FIG. **5**, as the mating terminal or tab **38** is inserted into the mating slot **52** of the contact portion **12** of the terminal **10**, a leading portion **58** of the mating terminal or tab **38** engages the embossment **40** of the second section or locking latch **26**. As the second section or locking latch **26** is fixed to the base portion **30**, the engagement of the mating terminal or tab **38** with the embossment **40** causes the embodiment **40** and the free end **34** of the second section or locking latch **26** to be elastically deformed, stressed or moved away from the mating terminal engagement surfaces **50** of the resilient arms **22***a*, **22***b*, as represented by arrow **60** in FIG. **5**. As the effective beam length D**1** of the second section or locking latch **26** is long, the second section or locking latch **26** does not take a permanent set as the movement occurs.
[0033] Insertion of the mating terminal or tab **38** into the mating slot **52** of the contact portion **12** of the terminal **10** is continued until the embossment **40** of the second section or locking latch **26** is positioned in a locking latch receiving opening **62** of the mating terminal or tab **38**, as shown in FIG. **6**. As the second section or locking latch **26** does not take a permanent set during insertion, the free end **34** and the second section or locking latch **26** are resiliently returned toward their unstressed position as the embossment **40** moves into alignment with the locking latch receiving opening **62** of the mating terminal or tab **38**. With the embossment **40** positioned in the locking latch receiving opening **62** of the mating terminal or tab **38**, the mating terminal or tab **38** is secured in the terminal **10** and the housing **82**.
[0034] If a mating terminal or tab **38** is to be removed from the terminal **10** and the housing **80**, the housing **80** is moved relative to the terminal **10** in the direction of **64** of FIG. **7**. As the user pulls backward on the housing **80**, the locking latch engaging surface **88** engages the free end **34** of the second section or locking latch **26**. As the movement of the housing **80** continues, the free end **34**

of the second section or locking latch **26** is cammed downward, in the direction of **66**. This continues until the housing **80** is positioned relative to the terminal **10** in the position shown in FIG. **7**. In this position, the embossment **40** is moved out of the locking latch receiving opening **62** of the mating terminal or tab **38**. With the embossment **40** moved out of the locking latch receiving opening **62** of the mating terminal or tab **38**, the mating terminal or tab **38** may be removed from the terminal **10** and the housing **82**.

[0035] As the second section or locking latch **26** is fixed to the base portion **30**, the free end **34** and the embossment **40** may be elastically deformed, stressed or moved away from the mating terminal or tab **38**, as represented by arrow **60** in FIG. **5**. As previously stated, the long effective beam length D**1** of the second section or locking latch **26**, the second section or locking latch **26** does not take a permanent set as the movement occurs.

[0036] The use of the second section or locking latch **26** with the free end **34** at the mating interface of the contact portion **12** of the terminal **10** allows the second section or locking latch **26** to be engaged from the mating interface rather than from the back of the contact portion **12**, as is done in the known art. Consequently, the terminal **10** of the present invention does not require a latch tip to extend from the back of the second section or locking latch **26** beyond the resilient contact arms **22***a*, **22***b*. As the latch tip of known terminals is stamped and formed from the base of the contact section of the terminal, As the latching tip is not needed, there is no need to stamp and form the latching tip from the base. Therefore, the height and length of the terminal **10** is minimized compared to known quick disconnect terminals. In addition, as the base portion **30** of the contact portion **12** of the terminal **10** does not have a cutout section, the current capacity is greater than known quick disconnect terminals which have a cutout section.

[0037] While the invention has been described with reference to a preferred embodiment, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the spirit and scope of the invention as defined in the accompanying claims. One skilled in the art will appreciate that the invention may be used with many modifications of structure, arrangement, proportions, sizes, materials and components and otherwise used in the practice of the invention, which are particularly adapted to specific environments and operative requirements without departing from the principles of the present invention. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive, the scope of the invention being defined by the appended claims, and not limited to the foregoing description or embodiments.

Claims

1. An electrical terminal comprising: a contact portion having a bottom wall with resilient contact arms extending therefrom; the bottom wall having a base portion proximate a back end of the bottom wall, the back end being removed from a mating interface of the bottom wall; a locking latch extending from and integrally attached to the base portion, the locking latch having a free end proximate the mating interface; wherein the free end of the locking latch is moved in a direction perpendicular to an axis of insertion as a mating terminal is inserted into the contact portion of the terminal.

2. The electrical terminal as recited in claim 1, wherein the locking latch has a securing section which is positioned in a plane which is above and essentially parallel to a plane of the base portion, the securing section engages the mating terminal as the mating terminal is inserted into the terminal.

3. The electrical terminal as recited in claim 2, wherein a projection is provided on the securing section of the locking latch.

4. The electrical terminal as recited in claim 3, wherein the projection is spaced from the base portion by a distance, the distance configured to provide an effective beam length which prevents

the locking latch from taking a permanent set as the mating terminal is inserted into the terminal.
5. The electrical terminal as recited in claim 2, wherein a bent section of the locking latch extends from the base portion.
6. The electrical terminal as recited in claim 5, wherein a sloping mating section is provided proximate the free end of the of the locking latch.
7. The electrical terminal as recited in claim 6, wherein the securing section extends between the mating section and the mounting section.
8. The electrical terminal as recited in claim 1, wherein the locking latch is separated from a first section of the bottom wall by a first slot and is separated from a third section of the bottom wall be a second slot, wherein the locking latch moves independently of the first section and the third section.
9. The electrical terminal as recited in claim 8, wherein the terminal has a first resilient contact arm of the resilient contact arms which extends from the first section of the bottom wall and a second resilient contact arm of the resilient contact arms which extends from the third section of the bottom wall.
10. The electrical terminal as recited in claim 9, wherein the first section and the third section extend from and are integrally attached to the base portion of the bottom wall.
11. The electrical terminal as recited in claim 10, wherein the first resilient arm and the second resilient arm have resilient contact sections with arcuate portions which extend from the bottom wall to mating terminal engaging members.
12. The electrical terminal as recited in claim 11, wherein mating terminal engagement surfaces are provided on the mating terminal engaging members, the mating terminal engagement surfaces are positioned at a top of a terminal mating slot.
13. A quick disconnect electrical terminal comprising: a contact portion having a bottom wall with resilient contact arms extending therefrom; the bottom wall having a base portion proximate a back end of the bottom wall, the back end being removed from a mating interface of the bottom wall, the bottom wall having a first section, a locking latch and a third section; the locking latch extending from and integrally attached to the base portion, the locking latch having a free end proximate the mating interface, the locking latch separated from the first section by a first slot and separated from the third section by a second slot; wherein the free end of the locking latch is moved independently of the first section and the third section in a direction perpendicular to an axis of insertion of a mating terminal as the mating terminal is inserted into the contact portion of the terminal.
14. The electrical terminal as recited in claim 13, wherein the locking latch has a securing section which is positioned in a plane which is above and essentially parallel to a plane of the base portion.
15. The electrical terminal as recited in claim 14, wherein a projection is provided on the securing section of the locking latch.
16. The electrical terminal as recited in claim 15, wherein the projection is spaced from the base portion by a distance, the distance configured to provide an effective beam length which prevents the locking latch from taking a permanent set as the mating terminal is inserted into the terminal.
17. The electrical terminal as recited in claim 16, wherein a bent section of the locking latch extends from the base portion.
18. The electrical terminal as recited in claim 17, wherein a sloping mating section is provided proximate the free end of the of the locking latch.
19. The electrical terminal as recited in claim 18, wherein the securing section extends between the mating section and the mounting section.
20. The electrical terminal as recited in claim 19, wherein the first section and the third section extend from and are integrally attached to the base portion of the bottom wall.